

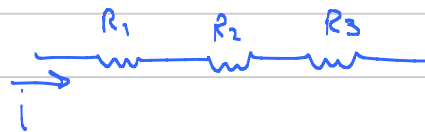


Definition of some terms

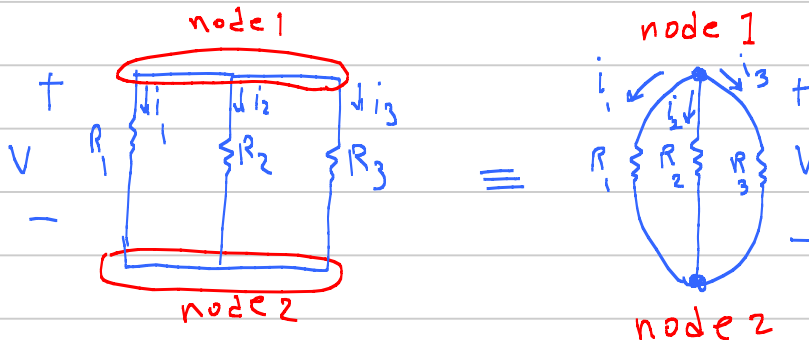
Node: The connection point between two or more circuit elements is called a node.

Loop: Any closed path in the circuit is called a loop.

Series connected elements: When two or more elements are connected sequentially, they have the same current and are said to be in series.



Parallel connected elements: When two or more elements are connected between the same node pair, they have the same voltage and are said to be in parallel.

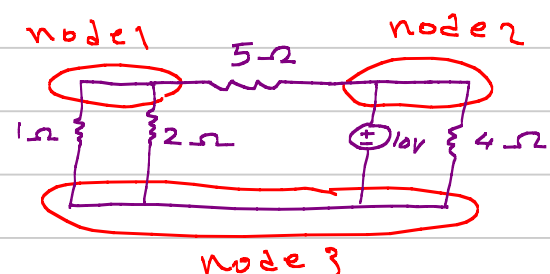


Ex. Identify the nodes, series elements and parallel elements.

3 nodes

$1\Omega \parallel 2\Omega$

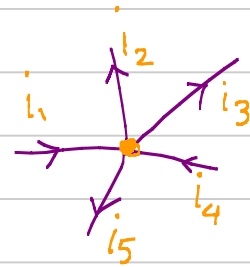
$10V \parallel 4\Omega$



No series connected elements

Kirchhoff's Current Law (KCL): The algebraic sum of all currents entering a node (or a closed boundary) is zero.

$$KCL: \sum_{n=1}^N i_n = 0$$

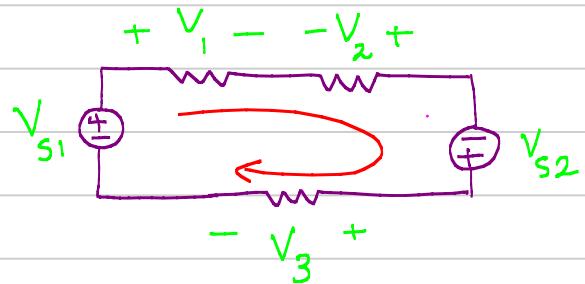


Taking the currents leaving the node positive.

$$KCL: -i_1 + i_2 + i_3 - i_4 + i_5 = 0$$

Kirchhoff's Voltage Law (KVL): The algebraic sum of all voltages around any loop in the circuit is zero.

$$\sum_{n=1}^N V_n = 0$$



Taking a voltage drop (from + to -) as a positive voltage:

$$KVL: V_1 - V_2 - V_{s2} + V_3 - V_{s1} = 0$$

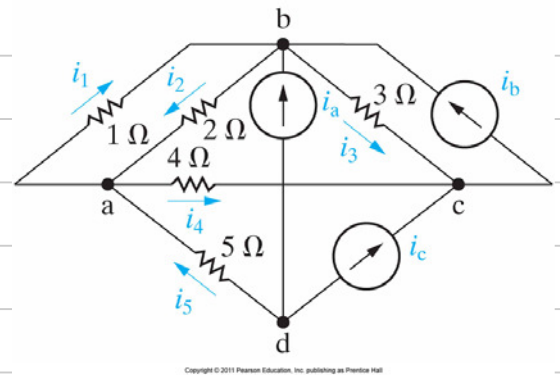
Ex. Write KCL at each node.

$$a: i_1 - i_2 + i_4 - i_5 = 0$$

$$b: -i_1 + i_2 - i_a + i_3 - i_b = 0$$

$$c: -i_c + i_b - i_3 - i_4 = 0$$

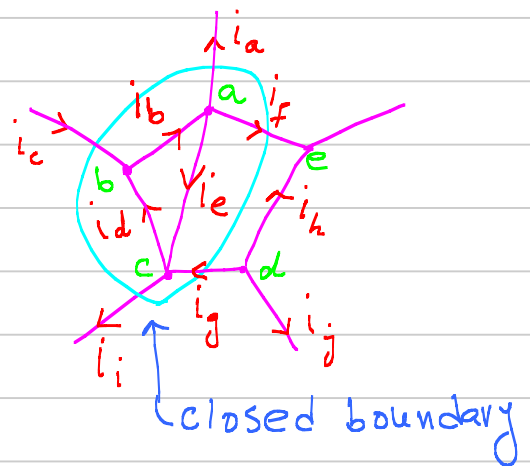
$$d: i_c + i_a + i_5 = 0$$



KCL at a closed boundary

$$\begin{cases} a: i_a + i_f - i_b - i_e = 0 \\ b: i_b - i_c - i_d = 0 \\ c: i_d - i_e - i_g + i_i = 0 \end{cases}$$

$$\text{Add: } i_a + i_f - i_c - i_g + i_i = 0 \quad (\text{KCL at closed boundary})$$



Ex. Use the basic laws to find i and check your answer using power balance.

KVL:

$$V_{ab} + V_{bc} - 120 = 0$$

Ohm's Law:

$$10i + 50i_1 - 120 = 0 \quad (1)$$

KCL at b :

$$-i + i_1 - 6 = 0 \quad (2)$$

From (1) & (2):

$$\underline{i = -3A}, \quad i_1 = 3A$$

check:

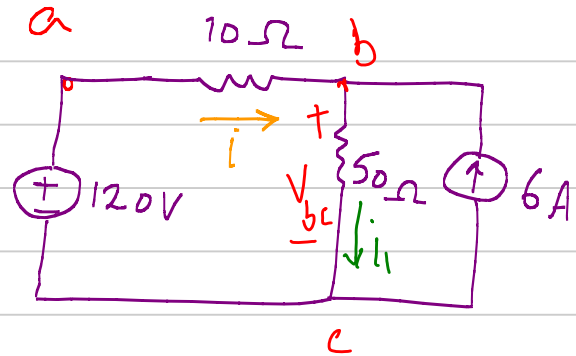
$$P_{120V} = -120(-3A) = 360W$$

$$P_{10\Omega} = Ri^2 = (10)(-3)^2 = 90W$$

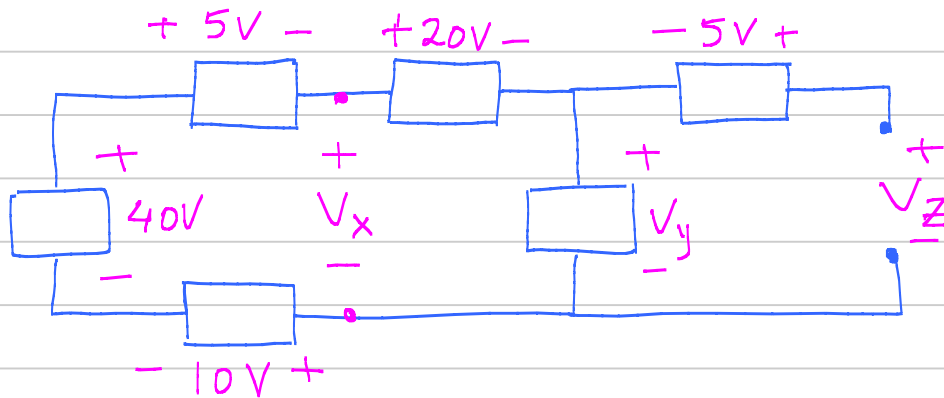
$$P_{50\Omega} = Ri_1^2 = (50)(3)^2 = 450W$$

$$P_{6A} = -(6)(V_{bc}) = -6(50 \times 3) = -900W$$

$$360 + 90 + 450 - 900 = 0 \quad \checkmark$$



Ex. Find V_x , V_y and V_z .



$$V_x = -5 + 40 - 10 = 25V$$

$$V_y = -20 - 5 + 40 - 10 = 5V$$

$$V_z = 5 + V_y = 5 + 5 = 10 \quad \leftarrow$$

$$\text{or: } V_z = -20 - 5 + 40 - 10 = 10 \quad \leftarrow$$

To find the voltage from a to b you can choose any path from a to b and add all voltage in that path.

Ex. Find V_o and I_o .

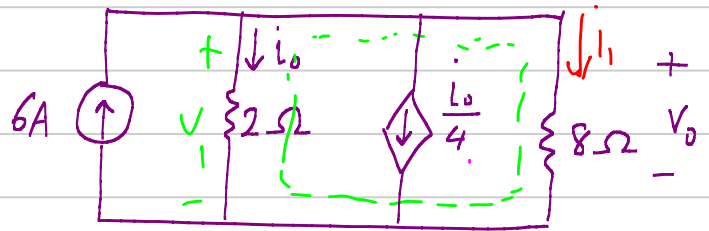
$$V_1 - V_o = 0$$

$$2I_o - 8I_1 = 0 \quad (1)$$

$$\text{KCL: } -6 + I_o + \frac{I_o}{4} + I_1 = 0 \quad (2)$$

$$(1), (2) \Rightarrow I_o = \underline{4A}, I_1 = 1A$$

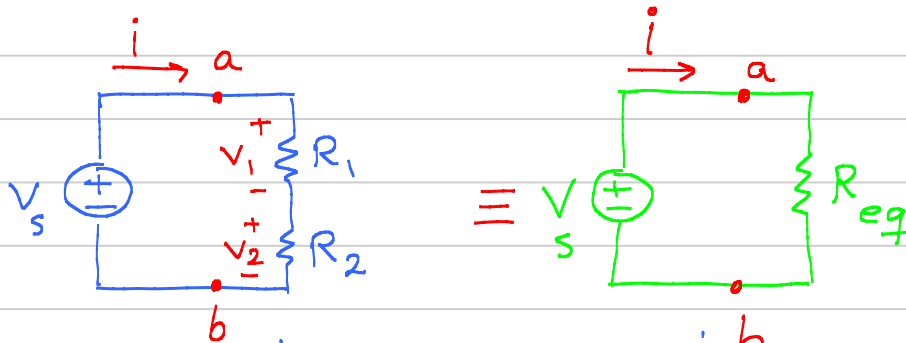
$$V_o = 8I_1 = \underline{8V}$$



Chapter 2, Resistive Circuits

Equivalent circuits: Two circuits are said to be equivalent if they have the same v-i characteristics at the specified pair of terminals.

Equivalent of series resistors:



$$\begin{cases} -V_s + R_1 i + R_2 i = 0 \Rightarrow i = \frac{V_s}{R_1 + R_2} \\ i = \frac{V_s}{R_{eq}} \end{cases}$$

$$\Rightarrow \frac{V_s}{R_1 + R_2} = \frac{V_s}{R_{eq}} \Rightarrow \boxed{R_{eq} = R_1 + R_2}$$

For N resistors in series :

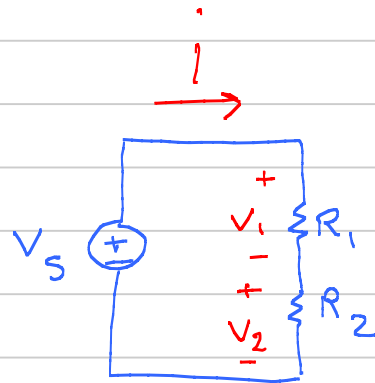
$$R_{eq} = R_1 + R_2 + \dots + R_N$$
$$= \sum_{i=1}^N R_i$$

Voltage divider rule :

$$i = \frac{V_S}{R_1 + R_2}$$

$$V_1 = R_1 i$$

$$= R_1 \frac{V_S}{R_1 + R_2}$$



$$V_1 = V_S \frac{R_1}{R_1 + R_2}$$

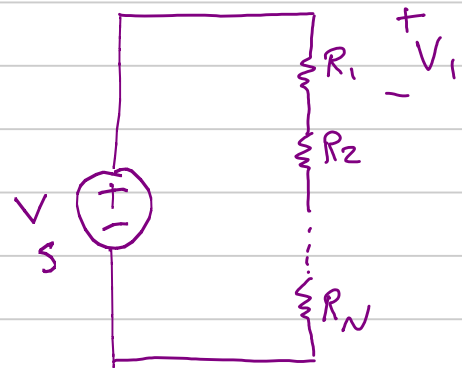
$$V_2 = V_S \frac{R_2}{R_1 + R_2}$$

with N resistors in series :

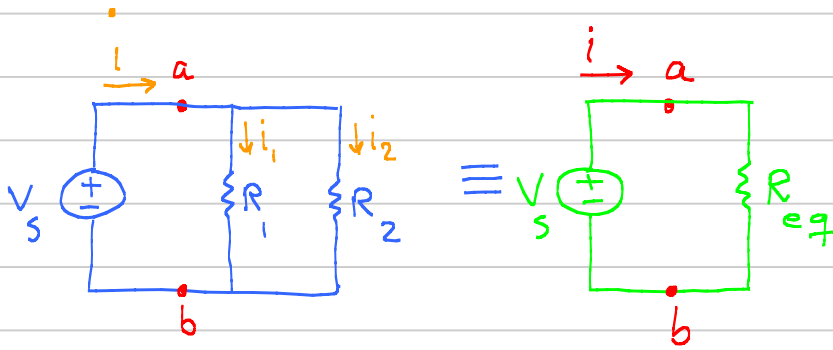
$$V_1 = V_S \frac{R_1}{R_1 + R_2 + \dots + R_N}$$

in general:

$$V_k = \frac{R_k}{\sum_{i=1}^N R_i} V_S$$



Equivalent of Parallel resistors:



$$i = i_1 + i_2 = \frac{V_s}{R_1} + \frac{V_s}{R_2} \quad i = \frac{V_s}{R_{eq}}$$

$$\Rightarrow \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R_{eq}}$$

$$\text{or, } R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

For N parallel resistors:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N} = \sum_{i=1}^N \frac{1}{R_i}$$

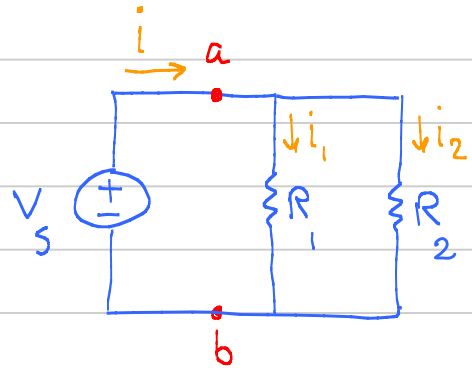
or, in terms of conductance:

$$G_{eq} = G_1 + G_2 + \dots + G_N = \sum_{i=1}^N G_i$$

Current divider rule:

$$V_s = R_{eq} i$$

$$V_s = \frac{R_1 R_2}{R_1 + R_2} i$$



$$i_1 = \frac{V_s}{R_1} \Rightarrow i_1 = \frac{R_2}{R_1 + R_2} i$$

$$i_2 = \frac{V_s}{R_2} \Rightarrow i_2 = \frac{R_1}{R_1 + R_2} i$$

in general:

$$i_k = i \frac{G_k}{\sum_{n=1}^N G_n}$$